

Level Sets and the Distribution of Zeros of Entire Functions

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[1]

Overview

This paper investigates a precise geometric mechanism by which translations of an even, real entire function produce new zeros on the imaginary axis. Specifically, for a function f drawn from a class $S_\infty(A)$ of even entire functions whose zeros are confined to a horizontal strip but which grow faster than any exponential, the authors prove that the combination

$$g_{t,\theta}(z) = e^{i\theta} f(z + t) + e^{-i\theta} f(z - t)$$

has **infinitely many zeros on the imaginary axis** for every nonzero real shift t and every phase $\theta \in \mathbb{R}$. The key insight linking these zeros to imaginary-axis behavior is that, when f is even, the values of $g_{t,\theta}$ on the imaginary axis reduce to $2 \operatorname{Re}(e^{i\theta} f(iy + t))$, which is exactly the level-set condition $\{\operatorname{Re}(e^{i\theta} f) = 0\}$ evaluated on a vertical translate — making the geometry of level curves the central analytic object.

Function Classes

The Strip Class $S(A)$

The paper operates inside the **strip class** $S(A)$, consisting of real entire functions of the Hadamard form[1]

$$f(z) = c e^{-az^2+bz} z^m \prod_{k=1}^{\infty} \left(1 - \frac{z}{z_k}\right) e^{z/z_k}, \quad a \geq 0, \quad z_k \in S(A) \setminus \{0\}.$$

The strip $S(A) = \{z \in \mathbb{C} : |\operatorname{Im}(z)| \leq A\}$ has width $2A$ and is symmetric about the real axis. The class $S(A)$ can be characterized intrinsically: $f \in S(A)$ if and only if f is the uniform limit on compact sets of real polynomials whose zeros all lie in $S(A)$ [1]. The Gauss–Lucas theorem implies the class is closed under differentiation. The boundary case $S(0)$ is the classical **Laguerre–Pólya class** $\mathcal{L}\text{-}\mathcal{P}$, the functions with only real zeros.

The Sub-class $S_{\infty}(A)$: Non-Exponential Type

The main results apply specifically to the sub-class $S_{\infty}(A)$, defined as those **even** functions in $S(A)$ that are **not of exponential type**[1]. An entire function is of exponential type if $|f(z)| \leq C_1 e^{C_2|z|}$ for some constants; this rules out all functions of order $\rho > 1$ and all functions of order 1 with unbounded zero-counting function $n(r)/r$. The role of the "even" and "non-exponential-type" hypotheses is sharp: the authors explicitly show that dropping either one makes the theorem fail.

- **Evenness is essential.** Without it the imaginary-axis identity $g_{t,\theta}(iy) = 2 \operatorname{Re}(e^{i\theta} f(iy + t))$ breaks down. The counterexample $f(x) = (x + 1)e^{-x^2}$ demonstrates that $g_{1/4,0}$ need not vanish on the imaginary axis for non-even f . [1]
- **Non-exponential type is essential.** The exponential-type function $f(z) = \cos(z)$ gives $g_{t,0}(z) = 2 \cos(z) \cos(t)$, which has only real zeros when $\cos(t) \neq 0$ — that is, $g_{t,0} \in S(A)$. This illustrates that exponential-type functions can avoid producing imaginary-axis zeros altogether. [1]

Because $f \in S_{\infty}(A)$ is even, its canonical product takes the symmetric form

$$f(z) = c e^{-az^2} z^{2m} \prod_{k=1}^{\infty} \left(1 - \frac{z^2}{z_k^2}\right),$$

with zeros paired at $\pm z_k$ [1]. The zero-counting function $n(r) = 2 \operatorname{card}\{k : |z_k| \leq r\} + 2m$ satisfies $n(r)/r \rightarrow \infty$, a key quantitative fact driving the whole argument.

The Main Theorem

Theorem 1.3 (Csordas–Smith). *Let $f \in S_\infty(A)$. Then for any $t \in \mathbb{R} \setminus \{0\}$ and any $\theta \in \mathbb{R}$, the entire function*

$$g_{t,\theta}(z) = e^{i\theta} f(z+t) + e^{-i\theta} f(z-t)$$

has infinitely many zeros on the imaginary axis.

The two most natural specializations are:

- $\theta = 0$: $g_{t,0}(z) = f(z+t) + f(z-t)$, the symmetric sum;
- $\theta = \pi/2$: $g_{t,\pi/2}(z) = i(f(z+t) - f(z-t))$, the antisymmetric difference.

In the language of infinite-order differential operators, these are $2 \cosh(tD_z)f(z)$ and $2i \sinh(tD_z)f(z)$ respectively, since $e^{tD_z}f(z) = f(z+t)$. [1]

Proof Architecture

The proof proceeds in three stages, each building on the previous.

Stage 1: Properties of the Logarithmic Derivative

Lemma 2.1 establishes the crucial quantitative unboundedness: defining the partial sum $\sigma(r) = \sum_{0 < \operatorname{Re} z_k \leq r} 1/|z_k|$, one has $\lim_{r \rightarrow \infty} [\sigma(r) - \sigma(r/2)] = \infty$ whenever $a = 0$ in the representation [1]. The proof is a combinatorial covering argument: if $\sigma(r) - \sigma(r/2)$ were bounded, a telescoping sum would force $n(r)/r$ to be bounded, contradicting the non-exponential-type assumption.

Lemma 2.2 converts this into estimates on the imaginary part of the logarithmic derivative f'/f . For $f \in S_\infty(A)$ and $\operatorname{Im}(z) > A$ one has $\operatorname{Im}(f'/f(z)) < 0$ — a strict negativity reflecting the fact that all

zeros lie below the line $\text{Im} = A$. Furthermore, in the vertical strip $V_t(r) = \{x + iy : 0 \leq x \leq t, y \geq r\}$, one has the lower bound[1]

$$\left| \text{Im} \frac{f'}{f}(z) \right| \geq 2a(r-1) + B[\sigma(r) - \sigma(r/2)], \quad z \in V_t(r-1) \setminus V_t(r)$$

where $B > 0$ is an absolute constant. This lower bound is unbounded in r , which is the engine of the whole argument.[1]

Stage 2: Level-Set Geometry via Schwarz's Lemma

Theorem 2.3 is a quantitative geometric result: if the disk $D(z_0, \delta(z_0))$ lies entirely in the set where $\text{Re}(e^{i\theta} f) > 0$ (or entirely < 0), then[1]

$$\delta(z_0) \leq 2 \left| \frac{f}{f'}(z_0) \right|.$$

The proof uses the Schwarz lemma via a composition $h = \varphi \circ (e^{i\theta} f) \circ \psi$ where ψ rescales and translates the unit disk into $D(z_0, \delta(z_0))$ and φ is a Möbius map sending the right half-plane to the unit disk. Since $h : D \rightarrow D$ with $h(0) = 0$, Schwarz gives $|h'(0)| \leq 1$, and unwinding the chain rule yields the stated bound[1]. In combination with Lemma 2.2, this means that for large r , the "gaps" between level curves are forced to become small — the level curves must cluster densely on horizontal segments.

Corollary 2.4 draws an immediate consequence: there exists some $t_0 \in (0, t]$ such that $g_{t_0, \theta}$ has at least one purely imaginary zero[1]. The argument is by contradiction: if $\text{Re}(e^{i\theta} f) \neq 0$ everywhere in $V_t(0)$, then the disk of radius $t/2$ around $t/2 + iy$ lies in a sign-definite region, bounding $|f'/f|$ from above by $4/t$ — directly contradicting Lemma 2.2's unbounded lower bound. This corollary is logically weaker than Theorem 1.3 but provides the first foothold: it guarantees existence of *at least one* imaginary-axis zero.

Stage 3: Counting Zeros via Tangent-Direction Analysis

The jump from "at least one" to "infinitely many" requires understanding the global geometry of level curves. Define the argument of the logarithmic derivative precisely: **Lemma 3.1** shows that for any $\varepsilon > 0$ and any zero $\pm iw$ with $|w| \geq M$, the angle $\text{Arg}(z/(z^2 - w^2))$ is within ε of $-\pi/2$ when $z \in V_t(M)$ with M large enough[1]. Geometrically, each term in the partial fraction expansion of f'/f contributes a direction that is nearly vertical.

Lemma 3.2 propagates this to the full logarithmic derivative: for large n and $z \in V_t(r_n - 1) \setminus V_t(r_n + 1)$, one has $|\text{Arg}(f'/f(z)) + \pi/2| < 2\varepsilon$ [1]. The proof assembles the partial fraction expansion, observes that finitely many terms become negligible for large $|z|$, and uses the convexity of the sector $\{w : |\text{Arg}(w) + \pi/2| < \varepsilon\}$ under addition.

Via the level-curve parameterization $e^{i\theta} f(z(s)) = is$, differentiating yields $z'(s) \cdot f'/f(z(s)) = 1/s$, so

$$\text{Arg}(z'(s)) = -\text{Arg} \frac{f'}{f}(z(s)).$$

Lemma 3.2 then says $\text{Arg}(z'(s)) \rightarrow \pi/2$ along the level curves in $V_t(r_n)$: **the tangents to the level curves become nearly vertical for large n** . Combined with the sign constraint from (2.5) that $\text{Arg}(z'(s)) \in (0, \pi)$ for $\text{Im}(z) > A$, this means that all level curves in this region are monotone — their tangents never point horizontally.[1]

Theorem 3.3 is the main counting theorem. Let $\ell(r; g_{t,\theta})$ count the zeros of $g_{t,\theta}$ on the imaginary segment $[0, ir]$, let $k_{t,\theta}(r)$ count level-curve crossings of the horizontal segment $[ir, t + ir]$, and let $j_{t,\theta}(y_1, y_2)$ count level-curve crossings of the vertical segment $[t + iy_1, t + iy_2]$. The monotonicity of level curves and the rectangle-counting argument yield the inequality[1]

$$k_{t,\theta}(r) \leq k_{t,\theta}(y_0) + j_{t,\theta}(y_0, r),$$

and the near-verticality of tangents (from Lemma 3.2) combined with Theorem 2.3 gives

$$k_{t,\theta}(r_n) \geq \frac{t}{4\Delta(r_n)} - 1, \quad \Delta(r_n) = \frac{2}{2a(r_n - 1) + B[\sigma(r_n) - \sigma(r_1)]}$$

Since $\Delta(r_n) \rightarrow 0$, the right side diverges. Translating back:

$j_{t,\theta}(y_0, r_n)$ counts the zeros of $g_{t,\theta}$ on $[iy_0, ir_n]$, and this count grows without bound. Theorem 1.3 follows immediately.[1]

The Role of Level Curves

Why Level Sets?

The central bridge between the zero distribution of f and the behavior of $g_{t,\theta}$ on the imaginary axis is the identity[1]

$$g_{t,\theta}(iy) = 2 \operatorname{Re}(e^{i\theta} f(iy + t)),$$

valid for all $y \in \mathbb{R}$ because f is even and real. A zero of $g_{t,\theta}$ on the imaginary axis is exactly a point where $t + iy$ lies on the level set $\{\operatorname{Re}(e^{i\theta} f) = 0\}$. The number of such zeros is therefore the number of times this level set crosses the vertical segment $\{t + iy : y \geq 0\}$. The entire proof becomes a study of how many times level curves thread back and forth across a given vertical strip.

Structural Properties of Level Curves

Level curves of an entire function are piecewise analytic curves. If z_0 is not a critical point of f (i.e., $f'(z_0) \neq 0$), then the level curve through z_0 is locally a smooth arc and f restricted to it is a local homeomorphism, so there is at most one zero of f on each unbranched arc. Branch points (critical points of f on the level set) can multiply zeros, but since $f \in S(A)$ implies $f' \in S(A)$, no branch points can occur outside $S(A)$. For even functions, all level curves have the twofold symmetry of the coordinate axes, allowing the analysis to concentrate on the first quadrant.[1]

Proposition 4.1: Interlacing of Level Curves

Proposition 4.1 shows that outside the strip $S(A)$, curves of $\{\operatorname{Re}(e^{i\theta} f) = 0\}$ interlace with those of $\{\operatorname{Re}(f) = 0\}$ [1]. More precisely, if $\operatorname{Re}(f(x_1 + iy_0)) = \operatorname{Re}(f(x_2 + iy_0)) = 0$ for $y_0 > A$,

then for any θ there exists $x_0 \in [x_1, x_2]$ with $\operatorname{Re}(e^{i\theta} f(x_0 + iy_0)) = 0$. The proof uses the fact that $\arg(f)$ is strictly monotone decreasing in the real direction for $\operatorname{Im}(z) > A$ (a consequence of $\partial_x \arg f = \operatorname{Im}(f'/f) < 0$), so $\theta + \pi/2 + \arg f$ sweeps through an interval of length π between the two zeros of $\operatorname{Re}(f)$, hence must pass through an integer multiple of π . This proposition implies that $|k_{t,\theta}(r) - k_{t,0}(r)| \leq 1$, i.e., the zero count is essentially independent of θ .

Connection to de Bruijn and the Riemann Hypothesis

The function class $S(A)$ and the shift operator $g_{t,\theta}$ connect directly to de Bruijn's 1950 work on trigonometric integrals[1]. De Bruijn proved that for the **imaginary** shift $g_{it,0}(z) = f(z + it) + f(z - it)$, the zeros move *toward* the real axis: if $f \in S(A)$, then $g_{it,0} \in S(\sqrt{A^2 - t^2})$ for $|t| < A$, and $g_{it,0} \in S(0) = \mathcal{L}\text{-}\mathcal{P}$ for $|t| \geq A$ [1]. This "zero attraction" phenomenon is precisely what underlies the de Bruijn–Newman constant and its role in the Riemann Hypothesis: the Riemann ξ -function is an element of $S(1/2)$, and its zeros are real iff $\xi \in \mathcal{L}\text{-}\mathcal{P}$.

The present paper addresses the **real** shift — a complementary and in some sense opposite regime. Where imaginary shifts compress zeros toward the real axis, real shifts of even non-exponential-type functions force infinitely many new zeros onto the *imaginary* axis. The contrast is sharp: for exponential type, real shifts can preserve the zero strip (as $\cos(z)$ demonstrates); for non-exponential type, they necessarily create imaginary-axis zeros.

Open Problems

The authors close with several open questions:[1]

Infinite-order operators. The operators $\cosh(tD_z)$ and $\sinh(tD_z)$ iterate to produce combinations like $(\cosh(tD_z))^2 f = f(z + 2t) + 2f(z) + f(z - 2t)$. It is open

whether these iterated combinations also have infinitely many imaginary-axis zeros for $f \in S_\infty(A)$.

The heat operator. The operator $\exp(t^2 D_z^2/2)f$ arises as $\lim_{n \rightarrow \infty} (\cosh(t D_z / \sqrt{n}))^n f$. Characterizing the functions in $S_\infty(A)$ for which this operator produces some nonreal zeros is described as "very difficult".[1]

Axis-confined zeros. Characterizing functions $f \in S_\infty(A)$ for which *all* zeros of $g_{t,0}$ lie on the coordinate axes is an open structural question.

Intersection with $\mathcal{L}\text{-}\mathcal{P}$. When $f \in S_\infty(A) \cap \mathcal{L}\text{-}\mathcal{P}$ (only real zeros, e.g., $f(z) = e^{-z^2}$), the function $g_{t,0}$ need not have nonreal zeros other than on the imaginary axis. Characterizing exactly when $g_{t,0}$ acquires such additional nonreal zeros remains open.[1]

Wave equation. Setting $w(x, t) = f(x + t) + f(x - t)$ for $x, t \in \mathbb{R}$, the function w satisfies the wave equation $w_{tt} = w_{xx}$, corresponding to two unit-speed waves traveling in opposite directions. This physical interpretation invites further study of the zero distribution from a PDE perspective.[1]

References

1. [LevelSetsAndDistributionsOfEntireFunctionZeros.pdf](#) - LEVEL SETS AND THE DISTRIBUTION OF ZEROS OF ENTIRE FUNCTIONS
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Abstract...